

Earning Roles: Self-Evolving Responsibility from Situated Peer Judgments

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Abstract

Multi-agent systems often assign work using fixed roles, global reputation, or terminal rewards. These signals can rank agents, but they cannot answer a sharper question: did this particular delivery help the particular agent that depended on it, and should that evidence change who receives the next task? We introduce Earning Roles, a self-evolving responsibility mechanism that turns a recipient’s use, revision, or rejection of a versioned delivery into evidence tied to the producer and task context. A responsibility check separates producer quality from recipient integration and downstream outcomes; later independent outcomes can revise the evidence before it informs another assignment. This gives each collaboration a local, attributable learning signal while allowing useful experience to transfer across recipients and tasks. We evaluate the mechanism on multi-agent tasks with genuine dependencies and delayed feedback, comparing it with static assignment, raw acceptance, contextual trust or bandit selection, terminal-only learning, and pooled history under matched information and cost. We measure whether it improves held-out task quality and total collaboration cost, while retaining prior competence under drift. These tests distinguish role learning from merely selecting a globally strong agent.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Cooperation and coordination*.

Keywords

multi-agent systems, self-evolving roles, peer judgment, online adaptation

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Internal pre-results manuscript. The scientific release gate is closed. All future result cells remain empty until an independently frozen experiment supplies the corresponding raw evidence.

1 Introduction

An agent can be useful in one collaboration and unsuitable for another. The reason is not only a latent capability: the recipient, dependency, artifact contract, and repair responsibility determine

whether a delivery helps. Existing multi-agent systems commonly use fixed roles, a global reputation, or a central selector. Such mechanisms can rank agents, but they do not establish whether a particular recipient used a particular delivery, whether the delivery caused the later action, or whether a reported repair was actually the producer’s responsibility.

This paper asks one question: *when can a real recipient’s situated judgment become reliable evidence for a producer’s future responsibility?* The question requires a complete causal chain:

delivery → recipient judgment and use → attributable role
evidence → pre-execution assignment → later quality and cost.

A label attached to a message, a final task score, or a change in a role posterior is not enough. A useful mechanism must preserve the recipient’s local information while keeping the producer, recipient, and final evaluator accountable for different parts of the outcome. This concern is related to reputation and partner-selection work [1–3], but the central object here is the attributable delivery and the later responsibility opportunity.

We contribute a protocol and a candidate update contract rather than an unsupported claim of a new optimizer. The protocol fixes selection before execution, selected-only delivery, delayed feedback, version binding, responsibility-aware labels, and complete cost accounting. The candidate update, called responsibility-aware role evidence (RARE), updates public evidence only when a recipient action can be attributed to the producer’s delivery. Raw acceptance, contextual trust/bandit selection, terminal-only feedback, and a pooled controller provide strong alternative explanations. The paper’s claim is conditional: RARE is useful only if it changes a later legal assignment and improves an unseen task stream beyond these matched alternatives.

The current evidence is deliberately limited. Two complete real-API development chains show that selection, recipient judgment, source snapshots, consumer execution, and a later assignment can be logged together. The same traces show that a four-check consumer score missed producer priority defects, that normal recipient integration was mistaken for an upstream repair, and that the feedback left the selection probability unchanged. These observations determine the next experiment; they are not a positive method result.

2 Problem and Information Contract

At decision t , x_t is the task context and C_t is the legal set of candidate producers. The selector samples a_t before any candidate executes:

$$a_t \sim \pi(\cdot \mid x_t, C_t, S_t), \quad a_t \in C_t, \quad (1)$$

where S_t is the public role-evidence state available at that time. Only a_t runs. The resulting delivery is o_t . The real owner or downstream recipient is j_t and must differ from a_t for a peer-judgment event.



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The recipient’s observation arrives after a delay d_t and is bound to the decision-time candidate version.

The event record is intentionally small:

$$e_t = (x_t, C_t, a_t, o_t, j_t, z_t, u_t, q_t, d_t, c_t), \quad (2)$$

where z_t is the recipient judgment, u_t is the recipient action, q_t is an independent producer-contract or final-quality observation, and c_t is the complete cost record. The judgment and action are distinct. We use the labels accept, integrate, upstream_fix, reject, and unknown; an unknown responsibility is missing information, not a negative label. The record also stores the behavior probability $p_t = \pi(a_t | x_t, C_t, S_t)$, artifact digest, candidate version, feature-schema version, and feedback watermark.

The online learner may read only events whose feedback has arrived. A hidden evaluator, future task score, or unexecuted candidate outcome cannot affect the current choice, retry, stopping rule, or update. Independent scores may be used after the stream for evaluation; they are not online labels unless the benchmark explicitly makes the check public and gives it to every condition.

3 Candidate Mechanism: RARE

The candidate mechanism maintains public evidence for producer v in context r as

$$R(v, r) = (n(v, r), q(v, r), u(v, r)). \quad (3)$$

Here n is the amount of eligible evidence, q is recipient evidence that can be attributed to the producer delivery, and u is evidence that is unresolved or rejected by an independent responsibility check. An event increments q only when the recipient actually inspected the delivery, performed a traceable action because of it, and the action is outside the recipient’s ordinary integration responsibility. Otherwise it increments u or remains unknown. This rule prevents a consumer’s own missing file or normal integration work from becoming a producer penalty.

Given a cached representation $h(x, v, r)$, the candidate score is

$$s(v | x, r) = w^\top h(x, v, r) + \beta \frac{q(v, r) - u(v, r)}{n(v, r) + \alpha}. \quad (4)$$

The selector uses a predeclared exploration floor and records its propensity. When a feedback event arrives, only the matching producer, context, and version receive one increment. Delayed events are combined using event-level sufficient statistics; if they cannot be combined without order dependence, the implementation replays the sealed event window and records the replay watermark. The old-task holdout rejects a write whose predeclared risk budget is exceeded.

RARE is a candidate mechanism, not evidence that this formula is optimal. In particular, the same event stream is given to a raw-acceptance rule and to a contextual trust or bandit baseline. If those baselines reproduce the future selections and utility, the paper will report that the responsibility-aware mechanism added no identifiable value. RLS, online logistic/SGD, and periodic refitting are training comparators; a small incremental state alone is not a novelty claim.

Condition	Information and purpose
Random/local uni-form	Candidate set only; selection and exploration lower bound.
No update	Initial prior and current task; tests whether learning is needed.
Raw acceptance	Recipient accept/reject without responsibility separation.
Contextual trust/bandit	Same context, judgment, propensity, and budget; strongest simple explanation.
Terminal-only	Independent final outcome without situated recipient evidence.
RARE	Responsibility-aware recipient evidence; candidate mechanism.
Pooled controller	All permitted public history under the same total resource budget.

Table 1: Predeclared comparison boundary. Extra close methods are added only when their implementation is faithful and their information and cost can be matched.

4 Benchmark and Evaluation Design

The working benchmark is a TeamBench-derived PeerRoleBench-TB candidate. Its qualification requires two structurally different task roots, an independently checked producer contract, a genuine recipient dependency, and a root-level development/confirmation split. The queue/priority DIST1 root is retained as a protocol diagnostic. The PIPE3 streaming root remains a conditional second candidate because a parent process can separately check producer output, recipient transformation, and downstream adoption. The current PIPE3 preflight is only a static contract/ledger fixture and listed-access canary; its task text also exposes the intended serialization bugs and therefore cannot support discovery or responsibility-attribution claims. A local adapter now constructs neutral task text and redacted public source payloads, but its runtime dispatch is unverified. The benchmark is not locked until task text, runtime payload/scorer isolation, and root-level split all qualify; renamed seeds are not independent task roots.

The first comparison table is fixed before confirmation:

We ask four research questions. RQ1 tests whether a recipient judgment predicts independent delivery and later-use outcomes beyond raw acceptance and terminal-only feedback. RQ2 tests whether public evidence changes a future assignment made before execution by a recipient that did not directly judge the candidate. RQ3 measures future prequential utility, full cost, update latency, state size, and old-task forgetting under selected-only delayed feedback. RQ4 tests responsibility ambiguity, recipient errors, version replacement, and task drift.

The primary unit of uncertainty is an independently reset stream. Episodes within a stream are dependent and are not treated as independent samples. The confirmation manifest fixes task roots, seeds, prompts, models, candidate menus, exploration, token and call budgets, failure handling, scorer versions, and the minimum useful effect before any confirmation output is observed. All real calls, tool actions, retries, UNKNOWNs, and wall-clock costs are streamed to auditable JSONL.

5 Current Evidence and Limitations

The completed v3 development run contains two limited real chains and six complete API responses. Both recipients declared a repair, both changed only their own integration file, and the recorded assignment was consumed by the next choice. The update mapping used the same 0.5 value as the prior, so the probability remained $[0.5, 0.5]$ and the selected peer did not change. The consumer score passed four checks but a separate diagnostic found that both producer priority artifacts violated the public constructor or import contract. These are protocol and scorer findings, not a role-learning measurement. The peers also shared a fresh model and prompt with no persistent private experience, so their identity labels cannot support a claim of learned specialization.

The consequence is a narrow current conclusion: the selected-only execution and audit plumbing is usable after responsibility and scorer repairs, while the mechanism’s signal, peer differentiation, and efficacy remain unestablished. No A800 training run is justified until a qualified task root produces an identifiable signal and a matched baseline contract.

6 Discussion and Falsification

The paper will withdraw the role-formation claim if a future assignment never changes, if the assignment is made only by the directly judging owner, if a same-information trust or bandit explains all gains, or if gains disappear at matched total cost. It will report a protocol or negative result if recipient judgments are not independently informative. It will not reinterpret a posterior change, a higher role entropy, or an isolated terminal success as evidence of useful role formation.

This boundary matters for related systems that learn reputations or partner choices [1, 3, 4]. The proposed question is not that agents can judge one another. It is whether a legally attributable, recipient-anchored judgment changes a later responsibility opportunity and improves a full collaboration objective under the same information and budget.

7 Conclusion

The current manuscript is a pre-results contract for a single empirical question. It defines the observation boundary, a responsibility-aware candidate update, a benchmark qualification path, and the strong explanations that must be defeated. The existing real traces justify repairing measurement and identity state; they do not justify claiming a working online role learner. The next scientific step is therefore a qualified second task root and a small, frozen, matched comparison—not a larger model or an unbounded training run.

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